Exhibit 4: Sample Completed Thinning Inspection Plot Form

THINNING INSPECTION PLOT FORM

page _1__of __1_

ID	(a) Forest *Green NF*	(b) District *Too Dense RD*	(c) Stand # *256-006*	(d) Inspector (Gov't) *I.M.Thin*	(e) Date *7/20/03*
Contract	(f) Contractor *I.M.Verigood*	(g) Sub-item # *1.1*	(h) Acres *20*	(i) % of Work Accomplished *100*	
	(j) Contract # *32789-00*	(k) Contract Inspector *F.M. Pay*		(l) % of Contract Time Left *10%*	

Contract Specifications	(m) Acceptable tpa	(n) Accept. Trees/plot	(o) Spacing	(p) Plot Size	(q) Min. Cut Ht	(r) Max. cut dbh	(s) Max. Cat C
	250 min/ *350* max *300* avg.	_*5*__min/ _*7*__max __*6*__ avg,	_*12*_ x _*12*___ feet	1:____ or 1/ *50*th acre	*24* in.	*5* in.	*15* tpa

Plot No. (1)	Tree Evaluation and Species (2): Code / Species; ~~Species~~ ? Tree was cut												SATISFACTORY: Satisfactory Leave Trees S1-S2	Surplus Trees S3-S4	Non-stocked points N1-N2	DEFICIENCY: Cat. "A" trees Improper Leave Tree Selection (Uncorrectable) A1-A2	A3-A4	Cat. "B" trees Excess (Correctable) B1-B2	B3-B4	Cat. "C" trees (Correctable) C1-C6
	1	2	3	4	5	6	7	8	9	10	11	12	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
1	S1	S1	S2	S2	S1	S1	S1	B1					7					1		
	LP	LP	PP	PP	LP	LP	LP	LP					Avg D.B.H. = 2.5 "; Avg HT = 15 ' Cat. C tally:							
Remarks:																				
2	S1	S2	S2	S2	S3	S1	S1	S1	C3				7	1						1
	LP	PP	LP	LP	LP	PP	PP	LP	LP				Avg D.B.H. = 3.0 "; Avg HT = 17 ' Cat. C tally: C3 = 1							
Remarks:																				
3	S1	S2	N1	N1	N1	N1							2		4					
	LP	PP											Avg D.B.H. = 3.5 "; Avg HT = 20 ' Cat. C tally:							
Remarks: *small meadow*																				
4	S1	S1	S1	S1	S1	N1							5		1					
	PP	PP	LP	LP	PP								Avg D.B.H. = 2.5 "; Avg HT = 15 ' Cat. C tally:							
Remarks:																				
5	S1	S1	A2	S1	S1	S1							5			1				
	PP	PP	~~PP~~	LP	PP	PP							Avg D.B.H. = 3.0 "; Avg HT = 12 ' Cat. C tally:							
Remarks:																				
6	S1	S2	B1	S1	S2	B2	S1						5					2		
	LP	PP	LP	PP	PP	LP	LP						Avg D.B.H. = 3.0 "; Avg HT = 15 ' Cat. C tally:							
Remarks:																				
7	S1	S2	S1	A1	N1								3		1	1				
	PP	PP	PP	~~PP~~									Avg D.B.H. = 3.0 "; Avg HT = 10 ' Cat. C tally:							
Remarks:																				
8	S1	S1	S1	S1	N2	S4	S4	S4					4	3	1					
	LP	DF	PP	LP		AS	AS	AS					Avg D.B.H. = 3.0 "; Avg HT = 12 ' Cat. C tally:							
Remarks:																				
9	S2	S2	S2	S2	S2	S2	S2	S3	S3				7	2						
	LP	PP	PP	LP	LP	PP	PP	PP	LP				Avg D.B.H. = 5.0 "; Avg HT = 25 ' Cat. C tally:							
Remarks:																				
10	S1	S1	S2	S1	S2	S2	C1						6							`1
	LP	LP	PP	PP	LP	PP	PP						Avg D.B.H. = 3.5 "; Avg HT = 10 ' Cat. C tally: C1 = 1							
Remarks:																				

(t) Thinning Quality Calculation		(3) Accept. Leave	(4) Surplus	(5) Non-stocked	(6) Cat A	(7) Cat A	(8) Cat B	(9) Cat B	(10) Cat C
	Totals of this page	51	6	7	2	0	3	0	2
	Total of all pages	51	6	7	2	0	3	0	2

1- Cat A (2 + 0) + Cat B (3 + 0) / Accept Leave (51) + Non-stocked (7) = .914= 91%

Cat C = 2 trees x (1/ 10 plots) x (50/1)=10 trees/ac

Submitted by: Igor M. Thin, Inspector 7/20/2003 All information is correct and accurate.

Exhibit 4: Sample Thinning Inspection Plot Form and Interpretations (Continued)

Note that the Contractor filled out the Inspection Sheet header completely. The Forest, District, Stand number, Sub-item, Acres, Contract # and all items in the Contract Specification row are provided by the Government either directly or within the Contract.

The Contractor provided the name of the Contractor, Contractor Inspector, % work accomplished, % contract time left, sheet number, date, and calculated the final quality and the Category C trees per acre. The Contractor assured that all columns and math were done correctly. The Contractor's Inspector signed and dated the inspection record and certified that the information was correct and accurate.

The plot information on the sample Inspection sheet was based on the following observations:

On each plot, the Contract required species to be recorded for each trees and an estimate of average leave tree dbh and average height. The Contractors observations are recorded in the second row of each plot.

Plot 1: This plot has seven satisfactory leave trees and one deficiency. Five satisfactory leave trees are within the thinning diameter range (S1), and two are above the maximum dbh (S2) and based on the prescription are included in the acceptable trees/plot determination. Additionally, a tree within the thinning range was left which should have been cut to meet the maximum allowable density (B1)

Plot 2: The plot has eight satisfactory leave trees and one deficiency. Four of the eight were satisfactory leave trees are in the thinning diameter range (S1) and four were larger than the maximum dbh. Since a maximum of seven satisfactory leave trees is allowed per plot, three trees are recorded as S2 and counted in the acceptable trees/plot calculation and one tree is recorded as S3 and considered surplus. Surplus trees, while satisfactory leave trees, are not counted in the acceptable trees/plot calculation. Additionally, one cut tree had a high stump (C3).

Plot 3: The plot has two satisfactory leave trees (S1, S2) and over half of the plot is a meadow with no trees. Three non-stocked points (N1) are recorded which brings the totals of S1, S2 and N1 to the minimum acceptable trees/plot.

Plot 4: This plot has five satisfactory leave trees (S1), and one large area, about 20% of plot where no trees exist (and none were cut) thus there is credit for one non-stocked points (N1). Had there not been an obvious void on the plot, then there would not have been credit for non-stocked points since the minimum acceptable tree/plot was met.

Plot 5: The plot has five satisfactory trees (S1), and one tree that was cut that was above the maximum dbh to cut and should not have been cut resulting in an A2 deficiency.

Plot 6: This plot has seven leave trees however only five are satisfactory leave trees. Of the leave trees, two were too close together (closer than the contract specification of 50% spacing variance) thus one tree is coded as a B1 deficiency, the other S1. Additionally, one tree was left with heavy mistletoe brooms and by contract specifications should not have been left; it is recorded as a B2 deficiency.

Plot 7: This plot has three satisfactory leave trees. There is a large void on the plot and the inspector found that one tree was cut that met leave tree specifications and was needed to meet the minimum trees/plot specification thus recorded one A1deficiencies. Additionally, one non-stocked point is recorded (N1) where no trees existed.

Plot 8: This plot has seven leave trees. Four were satisfactory leave trees in the diameter range for thinning (S1) and there was a clump of three aspen trees that based on the prescription were to be left (ignored) and not counted in the acceptable trees per plot calculation (S4). Additionally, one tree had a high level of mistletoe and was cut (appropriately) even though it created a "hole" in the plot; a credit N2 is recorded.

Plot 9: This plot had nine satisfactory trees on the plot because they were all above the maximum dbh to cut. Since the maximum density is seven trees per plot, only seven trees are recorded as S2 and counted in the trees per plot calculation based on the prescription and two are considered surplus trees (S3).

Plot 10: This plot has six satisfactory leave trees, three are in the range for thinning (S1) and three exceed the max. dbh to cut and based on the prescription are considered in the acceptable trees per plot calculation (S2). Additionally, one cut tree was not severed from the stump completely resulting in a C1 deficiency.

Calculate Totals.

$$\text{Quality \%} = 1.00 - \frac{\text{Cat A deficiencies (col. 6 + col. 7) + Cat B deficiencies (col. 8 + col. 9)}}{\text{Acceptable Leave (col. 3) + Non-stocked points (col. 5)}} \times 100$$

$$\text{Quality \%} = 1.00 - \frac{(2 + 0) + (3 + 0)}{51 + 7} \times 100 = 91.4\%$$

$$\text{Category C deficiencies per acre} = \frac{\text{Total Cat C (col.10)}}{\text{No. of plots}} \times \text{reciprocal of plot size}$$

$$\text{Cat. C deficiencies per acre} = \frac{2}{10} \times \frac{50}{1} = 10 \text{ trees/acre}$$

Since the Thinning Quality % is 90% or above AND the Category C deficiencies are below the maximum of 15 per acre allowed, the Contractor would receive full pay less other deductions.

Example of additional evaluations that can be conducted to evaluate if other conditions are met.

a. Total Satisfactory leave trees (S1 and S2) trees per acre

$$\frac{\text{Sum col 3}}{\text{No. of plots}} \times \text{reciprocal of plot size} = \frac{51}{10} \times 50 = 255 \text{ trees / ac}$$

b. Total of all trees per acre (excl. any small trees left on unit and trees below min. cut height).

$$\frac{\text{(Sum of col 3 + 4 + 6 + 7 + 8)}}{\text{No. of plots}} \times \text{reciprocal of plot size} = \frac{(51 + 6 + 2 + 0 + 3)}{10} \times 50 = 310 \text{ trees /ac}$$

Sample evaluation: The contract was developed to achieve a range of 250 to 350 trees per acre of desirable crop trees and included trees 5” dbh and greater (up to a maximum 350 tpa) in the range. Based on the above density calculations, the prescription goals were met. There are several natural openings in the stand and several desirable trees that were cut (“A trees”) that caused the total satisfactory leave trees per acre to be near the lower bounds of the acceptable level (355 vs. 350 minimum). The total of all trees left includes those left even if they were damaged (A4) or excess (B trees) and or surplus (S3, S4), resulting in 310 tpa, also within in the acceptable range, although it includes trees that are not desirable for future crop trees but are now part of the growing stock. Additional analysis should consider the distribution, species composition and other prescription objectives to determine if the overall thinning goals were met.